Rapport 15/10

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1 Sphere:

Let $n > 0$ and we consider the sphere parametrized by

$$S^{n-1} = \{x \in \mathbb{R}^n \quad | \quad ||x|| = 1\}$$

where the norm in this parametrization is the usual Euclidean norm. We can prove that S^{n-1} is an embedded submanifold of $\mathbb{R}^n$ with dimension $n-1$ thanks to the Definition 3.10 in the book "Introduction to optimization on smooth manifolds" of N. Boumal. It's easy to define a local defining function at each $x \in S^{n-1}$ as

$$h(x) = x_1^2 + x_2^2 + ... + x_n^2 - 1$$

It satisfies well conditions in the original condition: for any neighborhood $U \subset \mathbb{R}^n$ of x: $h(y) = 0 \iff y \in U \cap S^{n-1}$ and we have $Dh(x)[u] = 2\langle x, u\rangle$. Since $x \in S^{n-1}$, $x \neq 0$ then $\text{Rank}Dh(x) = 1$.

- **Tangent space:** With the local defining function above, for each $x \in S^{n-1}$, the tangent space at x is defined by

$$T_x S^{n-1} = \text{Ker}Dh(x) = \{u \in \text{R}^n | \langle x, u\rangle = 0\}$$

- **Riemannian gradient:** Let f be a smooth function from an embedded submanifold to $\mathbb{R}$, then its Riemannian gradient is a vector field on the tangent space. For the submanifold S^{n-1} embedded in $\mathbb{R}^n$ with the usual inner product which is also a valid Riemannian metric for S^{n-1} (Proposition 3.54). With this metric, based on the Proposition 3.61 in the book of Boumal, we can define a Riemannian gradient as follow: Let $\bar{f}$ be a smooth extension of f in a neighborhood of S^{n-1} (probaly in R^n) then

$$\text{grad}(f(x)) = \text{Proj}_x(\text{grad}\bar{f}(x))$$

  where $\text{Proj}_x(.)$ is the orthogonal projection into the tangent space at x. To find the orthogonal projection onto a space, suppose A with the basic $\{a_1, ..., a_k\}$, let S be the matrix whose columns correspond to each vector in the basic. Then the $\text{Proj}_A = S(S^TS)^{-1}S^T$.

In our problem, let $T_xS^{n-1} = \text{Ker}(x)$ then $(T_xS^{n-1})^\perp = \text{span}(x)$ then $\text{Proj}_{(T_xS^{n-1})^\perp} = x(x^Tx)^{-1}x^T = xx^T$(since $x \in S^{n-1}$). Then for $u \in \mathbb{R}^n$, we have

$$\text{Proj}_{T_xS^{n-1}}(u) = u - \text{Proj}_{(T_xS^{n-1})^\perp} = u - xx^Tu$$

This implies that $\text{grad}f(x) = (Id - xx^T)\text{grad}\bar{f}(x)$

- **Retraction:** R is a smooth map from the tangent bundle TS^{n-1} to S^{n-1} itself, i.e. $R(x, v) = R_x(v) \in S^{n-1}$ for $x \in S^{n-1}, v \in T_xS^{n-1}$. In an Euclidean space, $R_x(v) = x + v$, inspired by the Euclidean case, we can define a natural retraction for the sphere: we move from x along the direction v then project them to the sphere:

$$R_x(v) = \frac{x+v}{||x+v||} = \frac{x+v}{\sqrt{1+||v||^2}} \in S^{n-1}$$

Another choice used in the package **Pymanopt**:

$$R_x(v) = x.\cos(||v||) + v\frac{\sin(||v||)}{||v||}$$

Numerical example: We will tackle the following optimization problem:

$$\min_{x \in S^{n-1}} x^TAx$$

where A is a symmetric matrix. Define a smooth extension of f: $\bar{f}(x) = x^TAx$ on $\mathbb{R}^n$ then $\text{grad}\bar{f}(x) = 2Ax$, by the gradient computation above, the Riemannian gradient for f on S^{n-1} is $\text{grad}(x) = 2Ax - 2x(x^TAx) = 2(Ax - (x^TAx)x)$. Then $\text{grad}f(x) = 0$ iff x^TAx is an eigenvalue of A.

Theoretically, since A is symmetric, then its eigenvalues are all real and we also have

$$x^TAx \geq \lambda_{min}||x||^2 = \lambda_{min}$$

So $x^TAx = \lambda_{min}$ iff x is the eigenvector corresponding to the minimum eigenvalue of A. However, notice that x and $-x$ both satisfy this property. So if we try to run the optimization procedure for several initializations, we would obtain 2 points that minimize the loss function.

2 Symmetric definite matrices

Denote $\text{Sym}^+(n)$ and $\text{Sym}(n)$ be space of symmetric definite and symmetric (respectively) matrices. First observation, $\text{Sym}(n)$ is an Euclidean space equipped with the Frobenius inner product. And we could prove that $\text{Sym}^+(n)$ is an open subset of $\text{Sym}(n)$ implying that it's an open submanifold in $\text{Sym}(n)$ with same dimension of $\text{Sym}(n)$.

- **Tangent space:** Since it's an open submanifold of an Euclidean space, so its tangent space is $\text{Sym}(n)$, i.e. for each $x \in \text{Sym}^+(n)$, we have $T_x\text{Sym}^+(n) = \text{Sym}(n)$.

- **Riemannian gradient:** From now, Riemannian metric we will use is the Affine invariant (which is available in the package Pymanopt), at each $X \in \mathrm{Sym}^+(n)$, we define

$$\langle U, V\rangle_X = \mathrm{Tr}(X^{-1}UX^{-1}V), \forall U, V \in \mathrm{Sym}(n)$$

Let f be a smooth function from $\mathrm{Sym}^+(n)$ to $\mathbb{R}$ and $\bar{f}$ be an smooth extension of f in $\mathrm{Sym}(n)$. Then by Talor expansion for $X, H \in \mathrm{Sym}(n)$, we have

$$\bar{f}(X+H) = \bar{f}(X) + \langle \mathrm{grad}\bar{f}(X), H\rangle_F + O(H^2)$$

So the differential map of $\bar{f}$ is defined as

$$D\bar{f}(X)[H] = \langle \mathrm{grad}\bar{f}(x), H\rangle_F$$

By the Proposition 3.35, we have that if $X \in \mathrm{Sym}^+(n)$, we have

$$Df(X) = D\bar{f}(x)_{|T_X\mathrm{Sym}^+(n)} = D\bar{f}(x)$$

For all $H \in \mathrm{Sym}(n)$, we have $Df(X)[H] = \langle \mathrm{grad}\bar{f}(X), H\rangle_F = \mathrm{Tr}(\mathrm{grad}\bar{f}(X)H)$. So we define:

$$\mathrm{grad}f(X) = X\mathrm{grad}\bar{f}(X)X \implies Df(X)[H] = \langle \mathrm{grad}f(X), H\rangle_X$$

- **Retraction:** Recall that retraction $R_x(.)$ at $x \in \mathrm{Sym}^+(n)$ is a smooth function from $\mathrm{Sym}(n)$ to $\mathrm{Sym}^+(n)$. We will admit the following retraction (all the construction lies in the book of Boumal and the references therein):

$$R_X(V) = X^{1/2}\log(X^{-1/2}VX^{-1/2})X^{1/2}$$

Then with $X, X' \in \mathrm{Sym}^+(n)$, the distance associated to the corresponding retraction is

$$d(X', X)^2_{AI} = d(X, X')^2_{AI} = ||\log(X^{-1/2}X'X^{-1/2})||^2_F$$

where the function $\log : \mathrm{Sym}^+(n) \to \mathrm{Sym}(n)$ takes the logarithmic on the eigenvalues and the remains the eigenvectors of the SPD matrix.

Numerical example: We consider the problem of Karcher mean, that is finding

$$\arg\min_{A\in\mathrm{Sym}^+(n)} = \frac{1}{N}\sum_{i=1}^{N} d(A, A_i)^2$$

where $(A_i)_i$ is a set of random SPD matrices and $d(.,.)^2$ is the distance of 2 SPD matrices. We consider 2 cases:

- **Frobenius distance:** If $d(A, B) = ||A - B||^2_F$, then by using the convexity and Euclidean gradient method, we have:

$$A^* = \frac{1}{N}\sum_{i=1}^{N} A_i$$

which is also called **arithmetric mean**.

- **Affine invariant distance:** We consider $d(A,B) = ||\log(B^{-1/2}AB^{-1/2})||_F^2$. In this problem, we will proceed the Riemannian gradient descent:

$$X_0 \in \mathrm{Sym}^+(n), \quad X_{n+1} = R_{X_n}(-\eta_n \mathrm{grad} f(X_n))$$

Let $f(A) = ||\log(B^{-1/2}AB^{-1/2})||_F^2$ with $A \in \mathrm{Sym}(n)$, we will try to compute the $\mathrm{grad} f(A)$, let $H \in \mathrm{Sym}(n)$, we have that

$$\begin{aligned} f(A+H) &= ||\log(B^{-1/2}(A+H)B^{-1/2})||_F^2 = ||\log(B^{-1/2}AB^{-1/2} + B^{-1/2}HB^{-1/2})||_F^2 \\ &= ||\log(B^{-1/2}AB^{-1/2}) + \log(Id + B^{1/2}A^{-1}HB^{-1/2})||_F^2 \\ &= ||\log(B^{-1/2}AB^{-1/2}) + B^{1/2}A^{-1}HB^{-1/2} + O(H^2)||_F^2 \\ &= f(A) + 2\langle \log(B^{-1/2}AB^{-1/2}), B^{1/2}A^{-1}HB^{-1/2}\rangle_F + O(||H||_F^2) \\ &= f(A) + 2\langle A^{-1}B^{1/2}\log(B^{-1/2}AB^{-1/2})B^{-1/2}, H\rangle_F + O(||H||_F^2) \end{aligned}$$

Therefore, $\mathrm{grad} f(A) = 2A^{-1}B^{1/2}\log(B^{-1/2}AB^{-1/2})B^{-1/2}$, that implies that the Riemannian gradient of the Affine invariant distance is

$$\mathrm{grad}\bar{f}(A) = A\mathrm{grad} f(A)A = 2B^{1/2}\log(B^{-1/2}AB^{-1/2})B^{-1/2}A$$

Therefore, let $F(A) = \frac{1}{N}\sum_{i=1}^N d(A,A_i)^2$, then we have

$$\mathrm{grad} F(A) = \frac{1}{N}\sum_{i=1}^N 2A_i^{1/2}\log(A_i^{-1/2}AA_i^{-1/2})A_i^{-1/2}A$$

To check whether this gradient is true, we may use the `check_gradient` function in the package Pymanopt:

```
from pymanopt.tools import diagnostics
problem_kar = pymanopt.Problem(manifold_kar, cost_kar)
manifold_kar = positive_definite.SymmetricPositiveDefinite(n)

x = problem_kar.manifold.random_point()
d = grad_kar(x) ## grad_kar is the Riemannian gradient defined above
diagnostics.check_gradient(problem_kar, x, d)
```

The result is as in the figure below, the slope of the dashed line should match the slope of the continuous line.

Comments: Both arithmetric mean and Karcher mean are symmetric positive definites matrices. But the Karcher mean is found via an optimization procedure in the space of SPD matrices which isn't a linear space, whereas arithmetric mean is figured out in a usual Euclidean space. So if we take the Frobenius norm of the difference between 2 matrices, the gap could be very large.

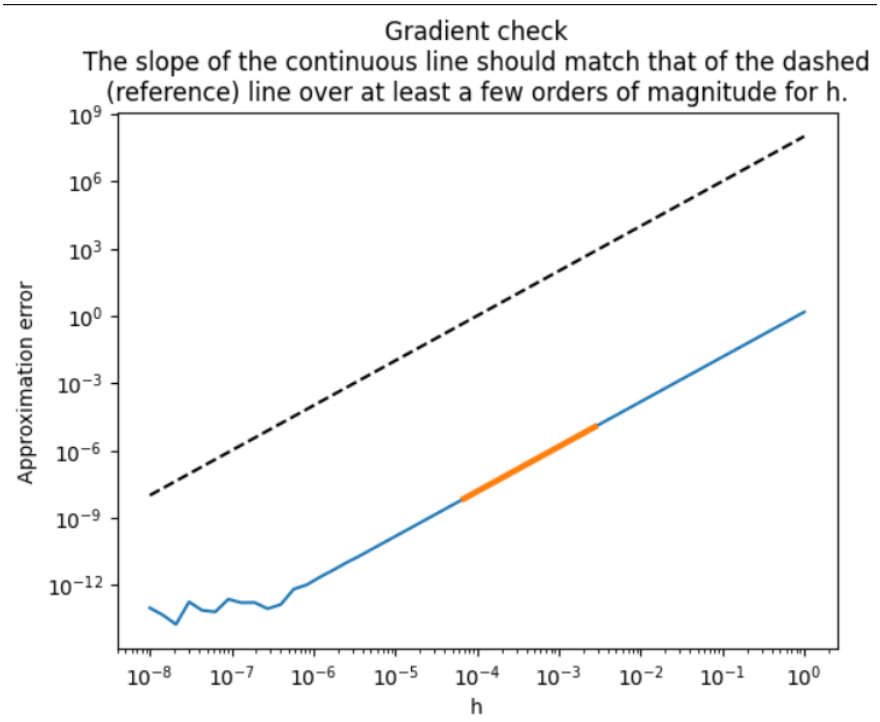


Figure 1: Gradient check